

Road Network Optimization of Intelligent Warehouse Picking Systems Based on Improved Genetic Algorithm

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Abstract: Intelligent Warehouse Picking System based on logistics robots is a new type of parts-to-picker order picking system, where robots carry mobile shelves to stationary pickers. The new picking mode puts forward higher requirements for the layout and design of warehousing network. In the existing few research on the path network optimization under the intelligent warehouse picking mode, the turning factors which obviously affects the picking efficiency, are seldom considered. In this paper, a mathematical model minimizing the total travel distance of logistics robots to complete all picking tasks is established, where the turning of robots is transformed into travel distance by cost function. Then an improved genetic algorithm with temperature parameter T and Metropolis acceptance criterion is proposed to solve the road network planning model. Finally, MATLAB is used to simulate and compare different road network layout strategies and algorithms from the total picking distance and total picking time to verify the effectiveness of the proposed method.

Keywords: Intelligent warehouse picking system; Path network; Genetic algorithm

1 Introduction

The intelligent warehouse system provides an effective scheme to solve the storage and logistics pressure caused by the rise of e-commerce. And the road network layout of intelligent warehouse system is the key to whether it can operate normally and improve its operation efficiency. In the practical application of intelligent warehousing and logistics robots, Kelly K. Bartlett [1] pointed out that the congestion of logistics robot system is closely related to the layout of path network, so the research on the road network layout of robots is a matter of concern to scholars. At the same time, the design of guidance path network of intelligent warehouse picking system has a great impact on the travel distance, space utilization and time spent on picking tasks of logistics robot, so the design of guidance path network is one of the key factors affecting the performance of logistics robot [2].

For the road network layout of intelligent warehouse, Tuan Le-Anh et al. [3] proposed that the road network layout usually includes unidirectional guidance path system, bidirectional guidance path system, single loop path system and series path system. Among them,

Gaskins et al. designed the unidirectional guidance path network by using the 0-1 integer programming method, and first proposed the shortest total distance of load operation as the design goal. Yan Weiguo [4] proposed a layout method of unidirectional guidance path network, aiming at the shortest distance and finding an optimal driving path for each logistics robot through Floyd algorithm. Wang Chuang, Chang Toyota [5] and others proposed to establish the equipment layout model of intelligent workshop based on the unidirectional guidance path of AGV, and use chaotic genetic algorithm to optimize the logistics path and solve the production equipment layout model. Wang Jianghua et al. [6] proposed a new layout optimization model of one-way and two-way hybrid path network, and then used the improved genetic algorithm to realize the path network planning. The unidirectional guidance path network is more suitable for the research of this paper.

The path network layout design problem belongs to NP complete problem [7-8]. Scholars at home and abroad use heuristic algorithms such as ant colony algorithm [9-10] and neighborhood search algorithm [11] to solve the large-scale path network layout design problem. Zhang Qi [12] used the improved ant colony algorithm to optimize the road network, and used the road resistance function to replace the geometric length of the road. Hu Kun [13] proposed an improved ant colony algorithm, which distributes pheromones according to whether there is congestion and different congestion conditions, so that the change of local path information of the current optimal path can be quickly reflected in the pheromone distribution of the path. Li Xiao et al. [14] proposed an improved ant colony algorithm, which added traffic congestion weight when initializing pheromone concentration to enhance the rationality of the path.

Doolan et al. [15] introduced a time ant algorithm based on ant colony optimization. He Chengwei, Mao Jian [16] constructed the MAKLINK undirected network graph, optimized the heuristic function in the ant colony algorithm, and constructed the obstacle avoidance factor to make the logistics robot path have the ability to avoid obstacles while reaching the global optimal path. Cao Xiaohua [17] proposed an improved particle swarm optimization (IPSO) algorithm suitable for collision avoidance decision optimization of multi-logistics robot system. Öncan T [18] proposed the branch and bound method to solve the problem of path network layout

design. Yuan Yang, Ye Feng [19] and others proposed a path planning method using the improved A * algorithm of load balancing to solve the problem of local congestion prevention and load balancing of multi logistics robot network in the current large-scale application scenario.

In summary, the existing academic research on the road network optimization is mostly under the traditional picking mode, but few research is under the intelligent warehouse picking mode. And the turning factor in the road network planning is rarely considered. When solving the mathematical model of the road network planning, the algorithms used are mainly swarm intelligence algorithms, such as ant colony algorithm and genetic algorithm, In this paper, in order to improve the picking efficiency of intelligent warehouse picking system, and reduce the travel distance of logistics robots, a model of minimizing the total travel distance is constructed, in which the A* algorithm considering turning factors is used to search the shortest path. And an improved genetic algorithm is designed to solve the model.

The rest of the paper is organized as follows. In Section 2 we describe the road network optimization problem in intelligent warehouse picking system. In Section 3 we build the mathematical model minimizing the total travel distance of logistics robots to complete picking tasks. In Section 4 we design an improved genetic algorithm to solve the model. In Section 5, we verify the proposed methods with a simulation model with real data. We finally conclude our study and discuss future research opportunities in Section 6.

2 Problem description

The path network optimization problem of intelligent warehouse picking system can be summarized as follows: given the guidance path network of the system, given handling tasks, it is required to determine the direction of each guidance path segment, ensure the strong connectivity of the designed unidirectional guidance path network, and optimize a certain performance of the system.

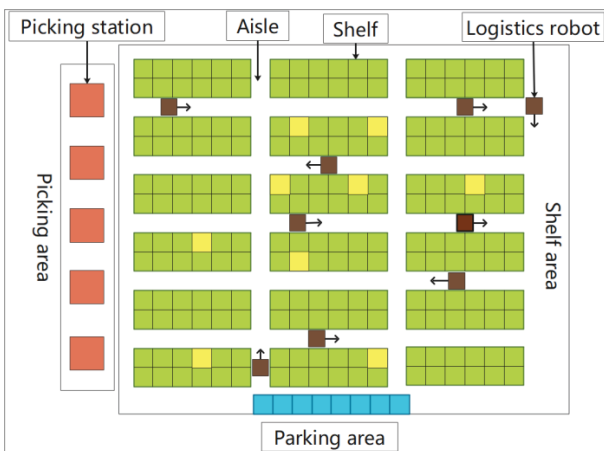


Figure 1 The typical layout of e-commerce intelligent warehouse

3 Model formulation

Evaluation criteria of path network: under different network layout strategies, logistics robots perform picking tasks according to the same path search rules, and determine the optimal solution of path network according to the total travel distance.

3.1 Model assumptions

According to the workflow of logistics robots when performing all picking tasks, the path network optimization problem puts forward the following assumptions:

- (1) Each shelf serves only one picking station for each handling;
- (2) The task of each picking station is known;
- (3) When the logistics robot puts back the shelf that has completed this task, it immediately obtains a new picking task, regardless of the docking point of the logistics robot;
- (4) The dwell time of the logistics robot on the picking station is 3 seconds, and the lifting, shelf placement and turning time of the logistics robot are all 1 second;
- (5) The logistics robot has sufficient power and does not have to be charged during the task;
- (6) Each logistics robot runs according to the planned route, and the deviation is ignored.

3.2 Parameters definition

In this paper, the model of unidirectional guidance path network is constructed using grid topology. Firstly, the path network of grid topology graph with K nodes is abstracted as an undirected graph $G = (V, E)$, where the node set $V = \{v_i\}$, the edge set is $E = \{v_i, v_j\}$, the edge corresponding to node m and node n is e_{mn} , the direction is Z_{mn} , and the direction of the initial edge is uncertain.

The following parameters are all in the path network layout state:

parameters	definition
S	the picking work area set
G	the shelf area set
B	the parking area set
b	the number of logistics robots($i=1,2,3,\dots,b$)
q	the number of picking tasks
v_{load}	The load running speed of logistics robots
v_{unload}	the no-load running speed of logistics robots
s	the picking station($i = 1,2,3, \dots, i \in B$)
g	the number of shelves($j=1,2,3,\dots, j \in G$)
f	the flow between points
L	the shortest path between points

3.3 Model building

In order to improve the picking efficiency of intelligent warehousing system. When modeling, consider the flow of each working link in the picking process of the robots

The mathematical model of minimizing travel distance is established to determine the driving direction of each road section. In equation (1), J represents the minimum total travel distance.

$$\min J = \sum_{\substack{1 \leq i \leq b \\ 1 \leq j \leq q \\ 1 \leq i \leq s \\ 1 \leq j \leq g}} (f_{b_i g_j} \cdot L_{b_i g_j} + f_{g_j s_i} \cdot L_{g_j s_i} + f_{s_i g_j} \cdot L_{s_i g_j} + f_{g_i g_j} \cdot L_{g_i g_j}) \cdot a_{b_i q_j} \quad (1)$$

$$Z_{mn} = \begin{cases} 1, & \text{Means from node } m \text{ to node } n \text{ and } m \neq n \\ 0, & \text{Otherwise} \end{cases} \quad (2)$$

$$a_{b_i q_j} = \begin{cases} 1, & \text{The } i\text{-th logistics robot completes the } j\text{-th task} \\ 0, & \text{Otherwise} \end{cases} \quad (3)$$

The formula (1) is expressed as minimizing the total travel distance. Which $f_{b_i g_j} \cdot L_{b_i g_j}$ represents the distance from the parking position of the logistics robot to the target shelf. $f_{g_j s_i} \cdot L_{g_j s_i}$ and $f_{s_i g_j} \cdot L_{s_i g_j}$ indicate the distance between the target shelf and the picking station in picking process. $f_{g_i g_j} \cdot L_{g_i g_j}$ represents the distance from the current shelf to the next target shelf when the next picking task is carried out. $L_{s_i g_j}$, $L_{g_j s_i}$, $L_{g_i g_j}$ need to meet the shortest path constraint, and the shortest distance can be obtained through the A* algorithm search. Considering that in the actual operation process, turning will increase the distance traveled by the system. Therefore, this article changes the evaluation index of the A* algorithm from $f(n) = g(n) + h(n)$ to $f(n) = g(n) + h(n) + \sum(s(\varphi_{n,n+1}) + \rho)$, where $f(n)$ represents the cost value from the starting point to the current search node and the sum of the estimated value from the current node to the target node. $g(n)$ represents the actual cost value from the starting point to the current node, which is a certain value. $h(n)$ represents the estimated cost value from the current node to the target point, which is generally estimated by methods such as Euclidean distance. $\varphi_{n,n+1}$ is the angle when turning, $s(\varphi_{n,n+1})$ is the cost function of the turning angle, and ρ is a constant. In addition, the designed unidirectional guide path network is strongly connected, that is, any two points can be reached in both directions, which is judged by the strong connectivity of the reachability matrix.

4 Algorithm design

By analyzing the existing research methods of road network planning, it is found that the algorithms used in solving the mathematical model of road network planning are mainly swarm intelligence algorithms, such as ant colony algorithm and genetic algorithm. Compared with ant colony algorithm, genetic algorithm has stronger global search ability, but genetic algorithm has the characteristics of premature convergence. In view of this defect of genetic algorithm, temperature parameter T and metropolis acceptance criterion are added in its iteration process.

(1) Temperature parameter T

The temperature parameter T of simulated annealing is used to control the iteration times of genetic algorithm, determine the initial temperature T_0 , the end temperature T_{end} ($T_0 > T_{end}$), and the cooling rate $\lambda_{\ll 1}$. After each iteration, $T = T_0 * \lambda$, until $T \leq T_{end}$, terminate the iteration process.

(2) Metropolis acceptance criteria

Adopting the metropolis acceptance criterion can retain the poor solution of the offspring with a certain probability, so as to avoid the premature convergence of the genetic algorithm to a certain extent.

The specific steps of the improved genetic algorithm are as follows:

Step 1. Chromosome coding design. Binary is used to encode the road section. 0 and 1 represent the direction of the road section in the grid topology map. According to the direction rules of up north, down south, left West, right east, north to south is 1, and vice versa is 0. West to East is 1, and vice versa is 0.

Step 2. Initial population generation. The strong connectivity discrimination method of reachability matrix is used to determine whether the chromosome coding is strongly connected. A kind of chromosome code represents a kind of route network layout situation. For the directed graph $G = (V, E)$ of each path network layout, where the node set $V = \{v_1, v_2, \dots, v_n\}$, the edge set is $E = \{v_i, v_j\}$, there is a corresponding directed graph n -order reachability matrix, which is defined as:

$$h_{ij} = \begin{cases} 1 & \text{when } v_i = v_j \text{ or } v_i \text{ to } v_j \text{ reach} \\ 0 & \text{other} \end{cases} \quad 1 \leq i, j \leq n \quad (4)$$

If and only if $\sum_{i=1}^n \sum_{j=1}^n h_{ij} = n^2$, that is, when the reachability matrix of a directed graph is a full 1 matrix, it represents the strong connection of the directed graph.

Step 3. Fitness function calculation. According to the direction of each unidirectional guidance path network, the A* algorithm is used to search and obtain the corresponding total travel distance J . Thus, the fitness of each individual is determined and sorted according to the size of fitness. The fitness function in this paper is:

$$F = \frac{1}{J} \quad (5)$$

Where J is the total travel distance for logistics robots to complete all picking tasks.

Step 4. Choice. The initial population is selected by roulette. The probability of each chromosome individual being selected is positively correlated with the fitness. The greater the fitness, the greater the probability of individual being selected. In each iteration, in order to ensure that the optimal solution is retained, the first n individuals with large fitness are selected to directly enter the next generation, and the other individuals are selected according to the roulette selection method.

Step 5. Cross. In this paper, the two-point crossing method is adopted. The crossing probability is $P_c = 0.9$.

Select two parent individuals N_h and N_f in the parent population and generate the random number ε between $[0, 1]$. If $\varepsilon \leq P_c$, participate in the crossover, and then randomly select two positions x and y in the individual coding to exchange the genes of N_h and N_f between x and y , so as to generate two new individuals.

Step 6. Variation. In this paper, double-point mutation is used. The mutation probability is $P_m = 0.1$. Two positions are randomly selected in the individual chromosome coding. The genes at these two positions are reversed.

Step 7. Metropolis acceptance criteria. The fitness of the newly generated offspring is calculated. If the fitness of the offspring is greater than that of the parent, the offspring is retained. Otherwise, it is retained with a certain probability according to the metropolis acceptance criteria. The retention probability is:

$$P = \begin{cases} 1 & \Delta f > 0 \\ e^{\frac{\Delta f}{T}} & \Delta f \leq 0 \end{cases} \quad (6)$$

Where $\Delta f = f_{new} - f_{old}$, T is the current temperature. If the fitness of the offspring is greater than that of parent, that is, when $\Delta f > 0$, the offspring will be retained. Otherwise, the offspring will be retained according to the probability P .

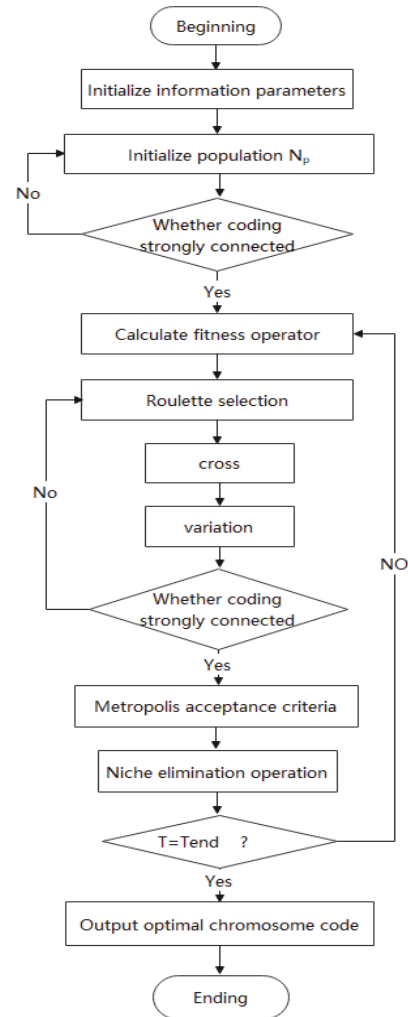
Step 8. Niche elimination calculation. In order to ensure the difference between individuals, the N individuals obtained through selection, crossover, and variation are combined with the N_0 optimal individuals directly retained from the parent. This is a total of $N + N_0$ individuals. Then the Hamming distance is calculated in pairs. For any two individuals, the Hamming distance shall not be less than the minimum value D_{min} . Hamming distance formula is expressed as:

$$D = \sum_{k=1}^m |x_{h_k} - x_{f_k}| \quad (7)$$

For the two individuals whose Hamming distance is lower than the minimum value, the individual with the smaller fitness is multiplied by the penalty factor ξ (value is 0.9). Then $N + N_0$ individuals are sorted according to the fitness. The top N optimal individuals are selected to enter the next generation.

Step 9. Termination conditions. Judge whether the current temperature T is less than T_{end} . If $T \leq T_{end}$, stop the iteration and output the result. Otherwise, continue selection, crossover, and variation until the iteration is completed.

The solution process of the improved genetic algorithm is shown in Figure 2.



5 Simulation analysis

In order to verify the effectiveness of the mathematical model and method of path network optimization of intelligent warehouse picking system proposed in this paper, simulation experiments are carried out using MATLAB.

5.1 Verification analysis

In order to facilitate the research of road network optimization, each edge in the warehouse is represented by nodes at both ends. The information of each channel in the warehouse is shown in Table I.

Table I Road section information of intelligent warehouse

Road section	node	node	distance
1	1	2	6
2	2	3	6
3	3	4	6
.....			
76	23	29	3
77	24	30	3
78	25	31	3
.....			

The number of sections in the warehouse is consistent

with the length of chromosomes, which is 113. One gene bit corresponds to the direction of one edge. The improved genetic algorithm proposed in this paper is used to solve the mathematical model of road network optimization.

The corresponding path network is shown in Figure 3:

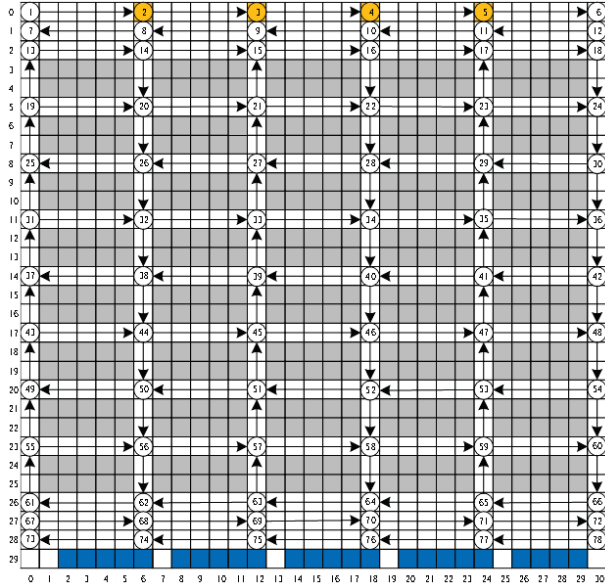


Figure 3 Optimal path network layout

In order to verify the effectiveness of the path network optimization algorithm proposed in this paper, different order quantities are selected. Firstly, the convergence of two different path network optimization methods, traditional genetic algorithm and improved genetic algorithm, are compared. It is verified that the improved genetic algorithm proposed in this paper can effectively avoid the premature convergence. Then the total travel distance and total picking time under different path network layout are compared. So as to verify the effectiveness of the proposed road network optimization model and improved algorithm.

5.2 The comparison of Algorithm performance

As can be seen from Figure 4, Under the scale of 1000 orders, the improved genetic algorithm reaches equilibrium at the 25th iteration, and the traditional genetic algorithm reaches equilibrium at the 16th iteration. The minimum total travel distance obtained by the traditional genetic algorithm is 130228cm. And the minimum total travel distance obtained by the improved genetic algorithm after multiple iterations is 107627cm. The quality of the optimal solution of the improved genetic algorithm is obviously better than that of the traditional genetic algorithm. Although the number of iterations of the improved genetic algorithm has increased in the running process, it can overcome the premature convergence characteristic of the traditional genetic algorithm, and has good optimization ability.

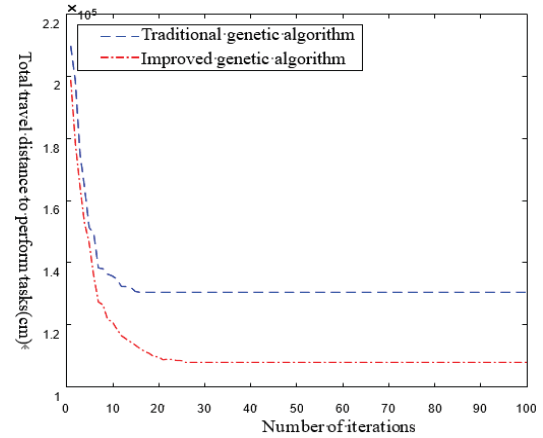


Figure 4 Comparison of optimization ability between traditional genetic algorithm and improved genetic algorithm

5.3 The comparison of total travel distance and total picking time

As shown in Figure 5 and Figure 6, with the increase of the number of orders, the total travel distance and total picking time of logistics robots performing all picking tasks gradually increases. But the total travel distance and total picking time of tasks performed by logistics robots in road network layout based on improved genetic algorithm is less than that in other road network layout. Therefore, the method proposed in this paper can effectively save picking time and improve picking efficiency.

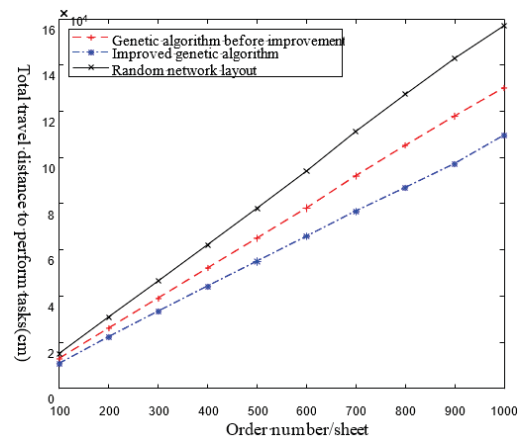


Figure 5 Comparison of the total travel distance to perform tasks under different road network planning strategies

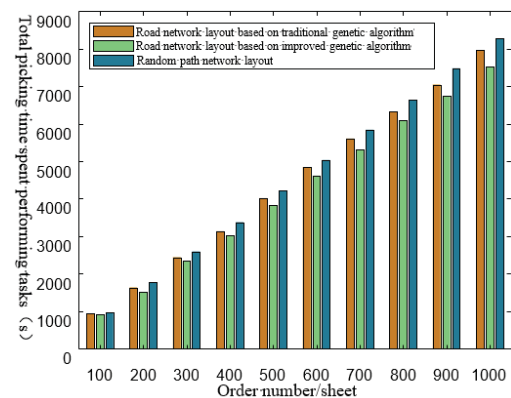


Figure 6 Comparison of the total picking time under different road network optimization strategies

6 Conclusions

This paper studies the path network layout optimization of the intelligent warehouse picking system. Firstly, the mathematical model of minimizing the total path of logistics robots to complete all picking tasks is established, where the turning factor is considered. Then the improved genetic algorithm is proposed to solve the model. Finally, the comparative experiments of traditional genetic algorithm, improved genetic algorithm and random three different road network layout strategies are designed. By comparing the total travel distance and total picking time of logistics robots under different layout strategy, the effectiveness of the mathematical model of road network optimization and the improved genetic algorithm proposed in this paper is verified.

Although this paper considers the characteristics of the actual picking operation of the logistics robot as much as possible, there are still some shortcomings. The calculation of the travel distance of logistics robots in this paper not considers the influence of dynamic factors on the travel distance in the real scenario. For example, the damage of logistics robots, the influence between logistics robots, and the emergency stop or failure in the handling process of logistics robots. In the future research, we will continue to conduct in-depth research on the impact of dynamic uncertainties on road network layout.

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